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II. Short Answer Questions

1. Explain how the definition of work in physics is different from general perception.
2. Write the various types of potential energy. Explain the formulae.
3. Write the differences between conservative and Non-conservative forces. Give two examples each.
4. Explain the characteristics of elastic and inelastic collision.
5. Define the following
   a) Coefficient of restitution
   b) Power
   c) Law of conservation of energy
   d) loss of kinetic energy in inelastic collision.

III. Long Answer Questions

1. Explain with graphs the difference between work done by a constant force and by a variable force.
2. State and explain work energy principle. Mention any three examples for it.
3. Arrive at an expression for power and velocity. Give some examples for the same.
4. Arrive at an expression for elastic collision in one dimension and discuss various cases.
5. What is inelastic collision? In which way it is different from elastic collision. Mention few examples in day to day life for inelastic collision.

IV. Numerical Problems

1. Calculate the work done by a force of 30 N in lifting a load of 2kg to a height of 10m ($g = 10 ms^{-2}$)

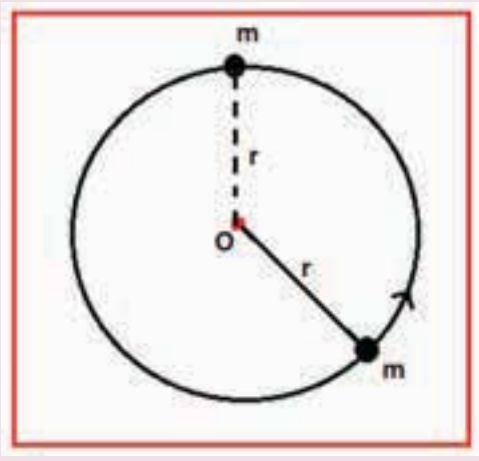
   Ans: 300J

2. A ball with a velocity of 5 ms^{-1} impinges at angle of 60° with the vertical on a smooth horizontal plane. If the coefficient of restitution is 0.5, find the velocity and direction after the impact.

   Ans: $v = 0.3\ m\ s^{-1}$

3. A bob of mass m is attached to one end of the rod of negligible mass and length r, the other end of which is pivoted freely at a fixed center O as shown in the figure. What initial speed must be given to the object to reach the top of the circle? (Hint: Use law of conservation of energy). Is this speed less or greater than speed obtained in the section 4.2.9?

   Ans: $v = \sqrt{4gr}\ m\ s^{-1}$



4. Two different unknown masses A and B collide. A is initially at rest when B has a speed v. After collision B has a speed v/2 and moves at right angles to its original